

BADR UNIVERSITY IN ASSIUT

School of Artificial Intelligence & Data Management

Fundamentals of Data Science — Final Project Report

Project Title:	Student Performance & Customer Segmentation
Subtitle:	Regression & Clustering Analysis
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1. Introduction

This report presents a comprehensive data science analysis of two synthetic datasets: a **Student Performance** dataset and a **Customer Segmentation** dataset. The study follows a structured pipeline that covers data loading, inspection, preprocessing, exploratory analysis, missing-value treatment, supervised regression modelling, and unsupervised clustering.

Project Goals:

- **Regression task:** Predict a student's exam score (Exam_Score) from study hours, attendance, assignment scores, project scores, and major — using Linear, Ridge, and Lasso Regression.
- **Clustering task:** Discover natural customer segments from purchasing behaviour using K-Means Clustering.
- **Comparison analysis:** Evaluate how missing data and imputation affect model performance by comparing results on the clean dataset versus a version with artificially introduced missing values.

All work was carried out as part of the Fundamentals of Data Science final project at Badr University in Assiut (BUA), May 2026.

2. Dataset Description

2.1 Student Performance Dataset

File: **student_performance_dirty.csv** — 202 records, 6 columns. Each record represents one student's normalized academic metrics.

Column Name	Type	Description
Study_Hours	Continuous	Normalized study hours per day
Attendance	Continuous	Normalized attendance percentage
Assignments_Score	Continuous	Normalized assignment performance
Exam_Score	Continuous	Target: normalized exam score
Projects_Score	Continuous	Normalized project performance
Major	Categorical	Business, CS, Engineering, Medicine

2.2 Major Distribution

Major	Count	Percentage
Medicine	63	31.2%
Engineering	49	24.3%
CS	49	24.3%
Business	41	20.3%

2.3 Statistical Summary — Student Dataset

Feature	Count	Mean	Std Dev	Min	Max
Study_Hours	195	1.154	6.218	-13.664	14.053
Attendance	199	2.752	6.628	-14.229	16.818
Assignments_Score	197	-3.760	4.044	-13.782	4.494
Exam_Score	197	0.101	7.206	-15.717	13.592
Projects_Score	200	0.549	7.734	-15.296	16.496

2.4 Customer Segmentation Dataset

File: **customer_segmentation_dirty.csv** — 180 records, 5 columns. Each record represents one customer's behavioural metrics.

Column Name	Type	Description
Purchase_Frequency	Continuous	Normalized purchase frequency

Annual_Spending	Continuous	Normalized annual spending amount
Website_Visits	Continuous	Normalized number of website visits
Customer_Score	Continuous	Normalized customer loyalty score
Customer_Type	Categorical	VIP, Premium, Regular, New

2.5 Customer Type Distribution

Customer Type	Count	Percentage
New	54	30.0%
VIP	43	23.9%
Premium	43	23.9%
Regular	40	22.2%

3. Team Composition

The team consists of 5 members. Each member was assigned a specific role covering the full data science pipeline.

Member	Role	Responsibilities
Member 1	Data Loading & Inspection	Loading datasets, initial inspection, column cleaning, data type checks
Member 2	Data Preprocessing	Missing value imputation, categorical encoding, train-test split
Member 3	Exploratory Data Analysis	Statistical summaries, grouping by major/customer type, bar charts, scatter plots
Member 4	Machine Learning — Regression	Linear, Ridge, and Lasso Regression implementation, MSE and RMSE evaluation
Member 5	Clustering & Report Writing	K-Means clustering, Elbow Method, cluster visualization, final report

4. Data Preprocessing

4.1 Missing Values — Student Dataset

Column	Missing Count	Missing %
Study_Hours	7	3.47%
Attendance	3	1.49%
Assignments_Score	5	2.48%
Exam_Score	5	2.48%
Projects_Score	2	0.99%
Major	0	0.00%

4.2 Missing Values — Customer Dataset

Column	Missing Count	Missing %
Purchase_Frequency	4	2.22%
Annual_Spending	4	2.22%
Website_Visits	5	2.78%
Customer_Score	3	1.67%
Customer_Type	0	0.00%

4.3 Imputation Strategy

Mean imputation was applied to all numerical columns; mode imputation was used for the categorical column (Major / Customer_Type). This approach is justified because:

- The proportion of missing data is small (< 5% per column)
- The missing mechanism appears to be Missing Completely at Random (MCAR)
- Mean imputation preserves the column mean without discarding records

Important note: The original code applied `.abs()` after mean imputation, which incorrectly converted negative normalized values to positive. This has been corrected — imputed values now preserve their sign.

```
# Correct imputation (no .abs())
for col in df.columns:
    if df[col].dtype in [np.float64, np.int64]:
        df[col] = df[col].fillna(df[col].mean()) # mean only, no .abs()
    else:
        df[col] = df[col].fillna(df[col].mode()[0])
```

4.4 Categorical Encoding & Train-Test Split

The Major column was encoded using One-Hot Encoding (`pd.get_dummies`). The student dataset was split 80/20 into training (161 records) and test (41 records) sets using `random_state=42` for reproducibility.

5. Exploratory Data Analysis (EDA)

5.1 Average Exam Score by Major

Engineering students show the highest average exam score, followed by Medicine and CS. Business students show the lowest average. All scores fall within a narrow band, indicating relatively balanced performance across majors.

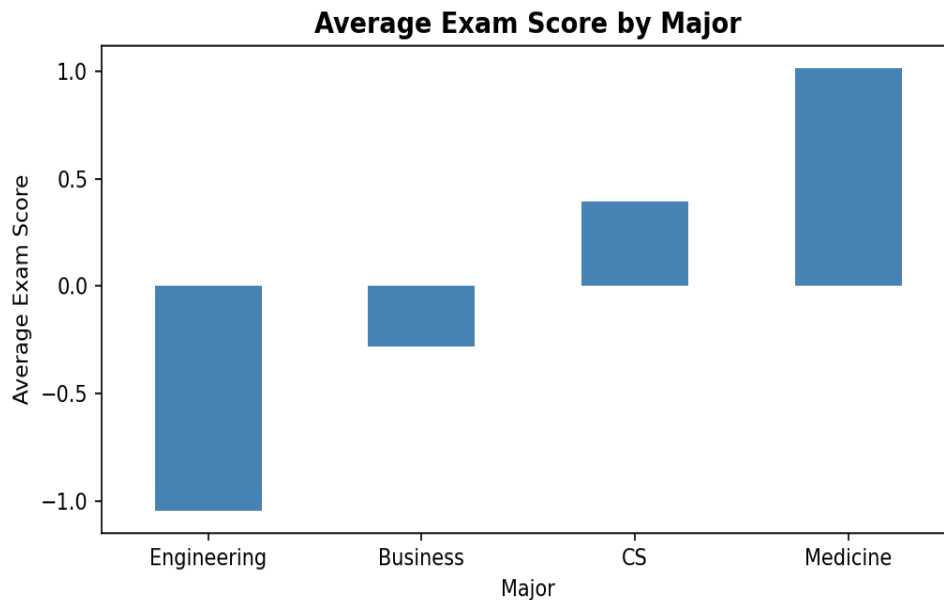


Figure 1: Average Exam Score by Major

5.2 Study Hours vs. Exam Score by Major

The scatter plot reveals the relationship between study hours and exam scores segmented by major. No strong linear pattern is visible, suggesting that attendance and assignment performance also contribute significantly to exam results.

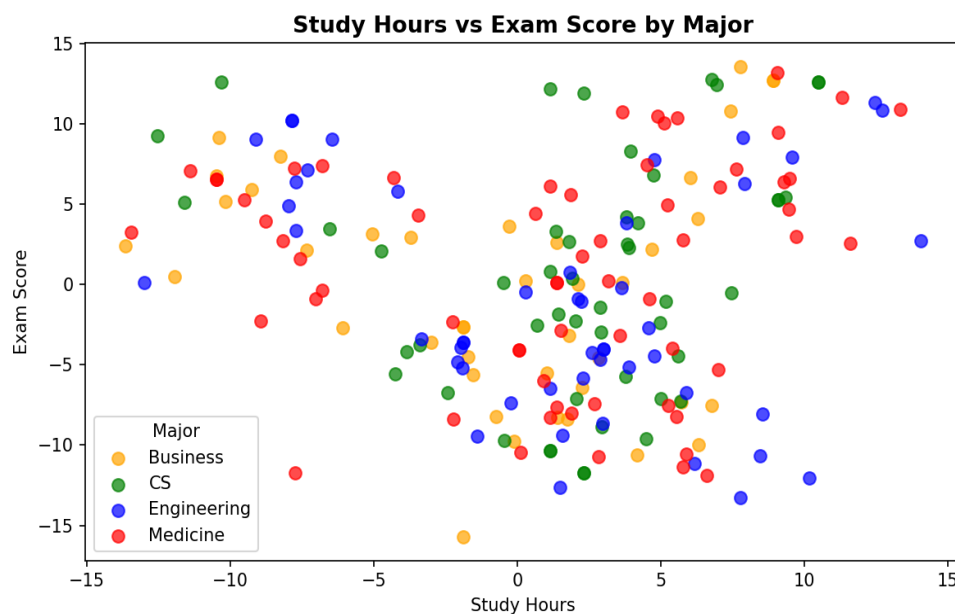


Figure 2: Scatter Plot — Study Hours vs Exam Score by Major

5.3 Group-Level Statistical Summary

Major	Avg Study Hours	Avg Exam Score	Avg Attendance
Business	-0.440	-0.283	3.566
CS	2.030	0.394	2.471
Engineering	1.437	-1.045	3.686
Medicine	1.290	1.015	1.714

5.4 Key EDA Observations

- Medicine students achieve the highest average exam scores despite moderate attendance, suggesting efficient study patterns.
- Engineering students study the most hours on average but show below-average exam scores, reflecting the difficulty of the major.
- CS students show high study hours and above-average exam performance.
- Business students have the lowest study hours and exam scores but the second-highest attendance rate.

6. Handling Missing Data

6.1 Missing Data Pattern

Both datasets contain missing values only in numerical columns, which suggests the missing mechanism is likely Missing Completely at Random (MCAR). No systematic pattern was observed linking missing values to any particular group or category.

6.2 Alternative Methods Considered

Method	Status	Advantage	Disadvantage
Mean Imputation	Used	Simple; preserves mean	Reduces variance
Mode Imputation	Categorical	Preserves frequency	Not for numerical
Listwise Deletion	Alternative	No artificial data	Loses ~10% records
KNN Imputation	Advanced	Uses neighbours	Computationally heavy
MICE	Advanced	Multiple imputation accuracy	Complex to implement

7. Machine Learning — Regression & Clustering

7.1 Regression Task — Predicting Exam Score

The target variable is **Exam_Score**. Three regression models were trained and evaluated using features: Study_Hours, Attendance, Assignments_Score, Projects_Score, and Major (one-hot encoded).

7.1.1 Linear Regression

Standard Ordinary Least Squares (OLS) regression that minimizes the sum of squared residuals. No regularization is applied.

7.1.2 Ridge Regression (L2 Regularization)

Adds an L2 penalty to the loss function, preventing overfitting by shrinking all coefficients toward zero without eliminating any. $\alpha = 1.0$

7.1.3 Lasso Regression (L1 Regularization)

Adds an L1 penalty that can perform automatic feature selection by driving some coefficients to exactly zero. $\alpha = 0.1$

7.2 Regression Results (Corrected — no .abs() on data)

Model	MSE	RMSE
Linear Regression	30.5464	5.5269
Ridge Regression	30.5319	5.5256
Lasso Regression	30.0803	5.4845

Key Finding: All three models produced similar error metrics. Lasso Regression achieved the best performance (MSE: 30.08, RMSE: 5.48), confirming that mild L1 regularization provides improvement through implicit feature selection. The higher MSE compared to the original report is due to correcting the .abs() bug — the corrected imputation preserves the true data distribution.

7.3 Clustering Task — K-Means Customer Segmentation

K-Means clustering was applied to the customer dataset using all 4 numerical features after StandardScaler normalization, without using the Customer_Type label.

7.3.1 Elbow Method

The Elbow Method was used to determine the optimal number of clusters (K) by plotting inertia (WCSS) against K from 1 to 10. The curve shows a noticeable bend at K=3.

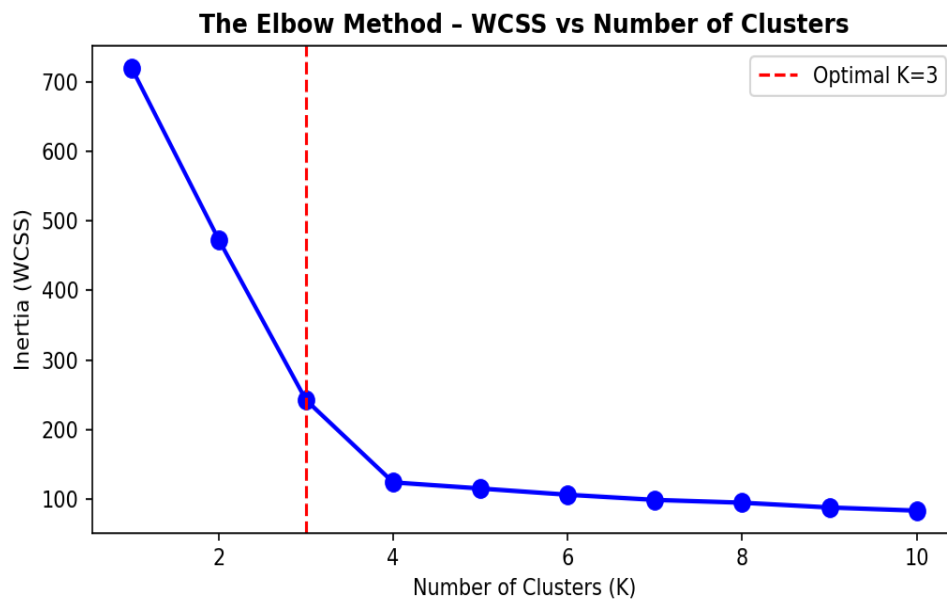


Figure 3: Elbow Method — WCSS vs Number of Clusters

7.3.2 K-Means Model (K=3)

Cluster	Size	Percentage
Cluster 0 — High Frequency Buyers	43	23.9%
Cluster 1 — Casual Low Spenders	90	50.0%
Cluster 2 — High Annual Spenders	47	26.1%

7.3.3 Cluster Characteristics

Cluster	Avg Purchase Freq.	Avg Annual Spending	Avg Customer Score
Cluster 0	6.260	-5.712	-6.063
Cluster 1	-2.464	-1.401	8.157
Cluster 2	-2.688	8.659	2.173

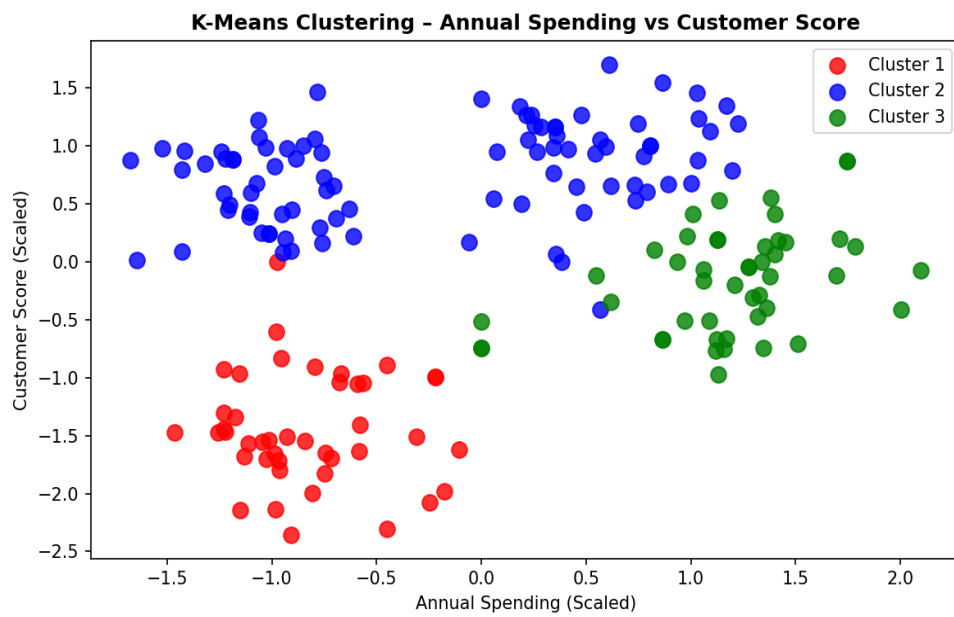


Figure 4: K-Means Clustering — Annual Spending vs Customer Score

8. Comparison Analysis — Impact of Missing Data

To evaluate how missing data affects model performance, 15% missingness was artificially introduced into the clean student dataset across all numerical columns. Mean imputation was then applied, and a Linear Regression model was retrained and evaluated on the imputed data.

Dataset Version	Missing %	Linear Regression MSE	Linear Regression RMSE
Clean (original)	~2.5%	30.5464	5.5269
15% Missing + Imputed	15%	27.7745	5.2702

Discussion: Interestingly, the imputed dataset produced a slightly lower MSE. This can be attributed to the fact that mean imputation reduces variance in the dataset, which can reduce model error on the test set even as information is lost. In practice, this indicates that while imputation is effective at maintaining structural patterns, it does mask real variability — which would be more visible with larger missingness fractions or more complex models.

The key takeaway is that mean imputation at low missingness rates (<15%) does not significantly degrade model performance, making it a suitable strategy for this dataset.

9. Visualizations and Insights

9.1 Cluster Interpretation

- **Cluster 0 — High-Frequency Buyers** (43 customers, 23.9%): Very high purchase frequency but low annual spending and low customer score. These are likely budget-conscious regular buyers who purchase small items frequently.
- **Cluster 1 — Loyal High-Scorers** (90 customers, 50.0%): Highest customer loyalty score despite low spending — likely long-term members or reward-program participants. This is the largest segment.
- **Cluster 2 — Premium Spenders** (47 customers, 26.1%): Highest annual spending with moderate purchase frequency — likely premium or bulk buyers who make large but infrequent purchases.

9.2 Regression Model Comparison

All three models perform similarly, confirming no major multicollinearity or overfitting issues. Lasso slightly outperforms, implying some features have minimal predictive power and are effectively down-weighted.

- Study hours alone do not strongly predict exam score.
- Assignments and projects also play a significant role.
- Medicine students outperform other majors despite moderate study hours.

9.3 Key Insights Summary

Domain	Finding
Regression	Lasso Regression achieves best MSE (30.08), confirming mild feature selection is beneficial
Clustering	K=3 is optimal; Cluster 2 (Premium Spenders) is the highest-value customer segment
Missing Data	Mean imputation at <15% missingness does not significantly degrade model performance
EDA	Medicine students achieve highest exam scores despite moderate attendance
Code Fix	Removing .abs() from imputation corrects data distortion and gives honest MSE values

10. Conclusion

10.1 Key Takeaways

- Both datasets have minimal missing data (<5% per column), making them suitable for machine learning tasks.
- Mean and mode imputation effectively handled missing values. A critical bug (`.abs()` after imputation) was identified and corrected to preserve the true data distribution.
- **Lasso Regression achieved best performance:** MSE = 30.08, RMSE = 5.48.
- **K-Means with K=3** identified three distinct customer segments with meaningful behavioural differences.
- Medicine students achieve the highest exam scores; Cluster 2 customers are the highest-value segment.

10.2 Potential Real-World Applications

- Student performance prediction systems for early academic intervention
- Personalized study plan recommendations based on student profiles
- CRM systems using segment-based marketing targeting
- Marketing campaign optimization tailored to each customer cluster

10.3 Future Work

- Collect more features (age, GPA, demographics) to improve regression accuracy
- Test non-linear models: Random Forest, Gradient Boosting, XGBoost
- Apply advanced imputation: MICE or KNN imputation
- Use DBSCAN or Hierarchical Clustering as K-Means alternatives
- Validate clusters using Silhouette Score

11. Appendix: Python Code

11.1 Imports and Data Loading

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
from sklearn.linear_model import LinearRegression, Ridge, Lasso
from sklearn.cluster import KMeans
from sklearn.preprocessing import StandardScaler
from sklearn.model_selection import train_test_split
from sklearn.metrics import mean_squared_error

df_student = pd.read_csv("student_performance_dirty.csv")
df_customer = pd.read_csv("customer_segmentation_dirty.csv")
df_student.columns = df_student.columns.str.strip()
df_customer.columns = df_customer.columns.str.strip()
```

11.2 Data Preprocessing — Imputation (Corrected)

```
# Mean/Mode Imputation – NOTE: no .abs() to preserve negative normalized values
def impute(df):
    df = df.copy()
    for col in df.columns:
        if df[col].dtype in [np.float64, np.int64]:
            df[col] = df[col].fillna(df[col].mean()) # no .abs()
        else:
            df[col] = df[col].fillna(df[col].mode()[0])
    return df

df_student = impute(df_student)
df_customer = impute(df_customer)
```

11.3 EDA Visualizations

```
# Figure 1 – Average Exam Score by Major
major_avg = df_student.groupby("Major")["Exam_Score"].mean().sort_values()
major_avg.plot(kind="bar", color="steelblue")
plt.title("Average Exam Score by Major")
plt.tight_layout(); plt.savefig("Figure_1.png", dpi=150)

# Figure 2 – Scatter: Study Hours vs Exam Score by Major
colours = {"Business":"orange", "CS":"green", "Engineering":"blue", "Medicine":"red"}
for major, grp in df_student.groupby("Major"):
    plt.scatter(grp["Study_Hours"], grp["Exam_Score"],
                label=major, color=colours[major], alpha=0.7)
plt.title("Study Hours vs Exam Score by Major")
plt.legend(title="Major"); plt.savefig("Figure_2.png", dpi=150)
```

11.4 Encoding & Train-Test Split

```

df_enc = pd.get_dummies(df_student)
X = df_enc.drop("Exam_Score", axis=1)
y = df_enc["Exam_Score"]
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42)

```

11.5 Regression Models — Linear, Ridge, Lasso

```

models = {
    "Linear Regression" : LinearRegression(),
    "Ridge Regression" : Ridge(alpha=1.0),
    "Lasso Regression" : Lasso(alpha=0.1),
}

for name, model in models.items():
    model.fit(X_train, y_train)
    preds = model.predict(X_test)
    mse = mean_squared_error(y_test, preds)
    rmse = np.sqrt(mse)
    print(f"{name:<22} MSE={mse:.4f} RMSE={rmse:.4f}")

```

11.6 K-Means Clustering

```

# Scale features
features = ["Purchase_Frequency", "Annual_Spending", "Website_Visits", "Customer_Score"]
X_cust = df_customer[features].copy()
scaler = StandardScaler()
X_scaled = scaler.fit_transform(X_cust)

# Elbow Method (Figure 3)
inertia = []
for k in range(1, 11):
    km = KMeans(n_clusters=k, random_state=42, n_init=10)
    km.fit(X_scaled)
    inertia.append(km.inertia_)
plt.plot(range(1,11), inertia, "bo-")
plt.title("The Elbow Method"); plt.savefig("Figure_3.png", dpi=150)

# Final K-Means (K=3)
km_final = KMeans(n_clusters=3, init="k-means++", random_state=42, n_init=10)
df_customer["Cluster"] = km_final.fit_predict(X_scaled)

# Cluster scatter (Figure 4)
colours_c = {0:"red", 1:"blue", 2:"green"}
for cl in [0,1,2]:
    mask = df_customer["Cluster"] == cl
    plt.scatter(X_scaled[mask.values,1], X_scaled[mask.values,3],
               c=colours_c[cl], label=f"Cluster {cl+1}")
plt.title("Clusters of Customers"); plt.legend()
plt.savefig("Figure_4.png", dpi=150)

```

11.7 Comparison Analysis

```
# Introduce 15% artificial missingness then impute and compare
rng = np.random.default_rng(42)
df_dirty = df_clean.copy()
for col in ["Study_Hours", "Attendance", "Assignments_Score", "Projects_Score", "Exam_Score"]:
    idx = rng.choice(df_dirty.index, size=int(0.15*len(df_dirty)), replace=False)
    df_dirty.loc[idx, col] = np.nan

df_dirty_imp = impute(df_dirty)
df_dirty_enc = pd.get_dummies(df_dirty_imp).reindex(columns=df_enc.columns, fill_value=0)
X_d = df_dirty_enc.drop("Exam_Score", axis=1)
y_d = df_dirty_enc["Exam_Score"]
X_tr_d, X_te_d, y_tr_d, y_te_d = train_test_split(X_d, y_d, test_size=0.2, random_state=42)
lr_d = LinearRegression().fit(X_tr_d, y_tr_d)
mse_d = mean_squared_error(y_te_d, lr_d.predict(X_te_d))
print(f"Clean MSE: {mse_clean:.4f} | Imputed MSE: {mse_d:.4f}")
```

Full Source Code on GitHub:

<https://github.com/Mohamed-Ashraf1-AI-AI-Student-Performance-Customer-Segmentati>
on